

MSE-1 DEVOPS ASSESSMENT REPORT

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Branch- CSE 'C'

Repository - <https://github.com/Kanishka780/github-skills-challenge>

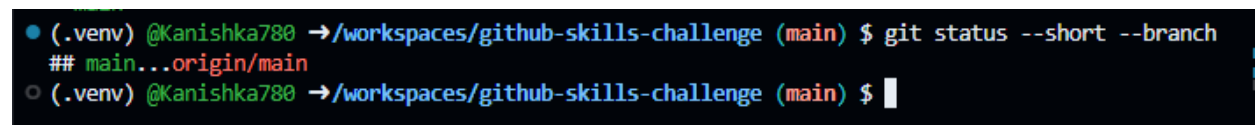
1. Objective

The objective of this assessment was to configure Continuous Integration for a Python application using GitHub Actions. The workflows were configured to run automatically when a pull request targets the `main` branch.

2. Development Environment

- Repository opened in GitHub Codespaces.
- Python virtual environment created and activated.
- Dependencies installed from `requirements.txt`.
- Testing tools installed:
 - `pytest`
 - `pytest-cov`
 - `coverage`
- Application source code and existing tests were not modified.

Heading: Development environment and project repository

A terminal window with a dark background and light blue text. The prompt is `(.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $`. The command `git status --short --branch` has been executed, resulting in the output `## main...origin/main`. The prompt is then `(.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $` with a cursor.

```
(.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $ git status --short --branch
## main...origin/main
(.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $
```

3. Test Execution

Result:

- Tests collected: 8
- Tests passed: 8
- Tests failed: 0

Heading: Successful Python test execution

```
● (.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $ source .venv/bin/activate
python -m pytest tests/ -v
===== test session starts =====
platform linux -- Python 3.13.15, pytest-8.4.1, pluggy-1.6.0 -- /workspaces/github-skills-challen
ge/.venv/bin/python
cachedir: .pytest_cache
rootdir: /workspaces/github-skills-challenge
plugins: cov-7.1.0
collected 8 items

tests/calculations_test.py::test_area_of_circle_positive_radius PASSED [ 12%]
tests/calculations_test.py::test_area_of_circle_zero_radius PASSED [ 25%]
tests/calculations_test.py::test_get_nth_fibonacci_zero PASSED [ 37%]
tests/calculations_test.py::test_get_nth_fibonacci_one PASSED [ 50%]
tests/test_aiops_pipeline.py::test_normal_record_is_not_anomaly PASSED [ 62%]
tests/test_aiops_pipeline.py::test_anomalous_record_is_detected PASSED [ 75%]
tests/test_aiops_pipeline.py::test_producer_publishes_event PASSED [ 87%]
tests/test_aiops_pipeline.py::test_consumer_receives_event PASSED [100%]

===== 8 passed in 0.09s =====
○ (.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $
```

4. Test Coverage

Result:

Total statements: 96

- Total coverage: 58%
- Coverage report generated successfully.
- Minimum enforced threshold: 50%

The coverage workflow fails if coverage falls below 50%.

Heading: Test coverage report

```
(.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $ python -m pytest tests/ -v --cov=src --cov-report=term-missing
plugins: cov-7.1.0
collected 8 items

tests/calculations_test.py::test_area_of_circle_positive_radius PASSED [ 12%]
tests/calculations_test.py::test_area_of_circle_zero_radius PASSED [ 25%]
tests/calculations_test.py::test_get_nth_fibonacci_zero PASSED [ 37%]
tests/calculations_test.py::test_get_nth_fibonacci_one PASSED [ 50%]
tests/test_aiops_pipeline.py::test_normal_record_is_not_anomaly PASSED [ 62%]
tests/test_aiops_pipeline.py::test_anomalous_record_is_detected PASSED [ 75%]
tests/test_aiops_pipeline.py::test_producer_publishes_event PASSED [ 87%]
tests/test_aiops_pipeline.py::test_consumer_receives_event PASSED [100%]

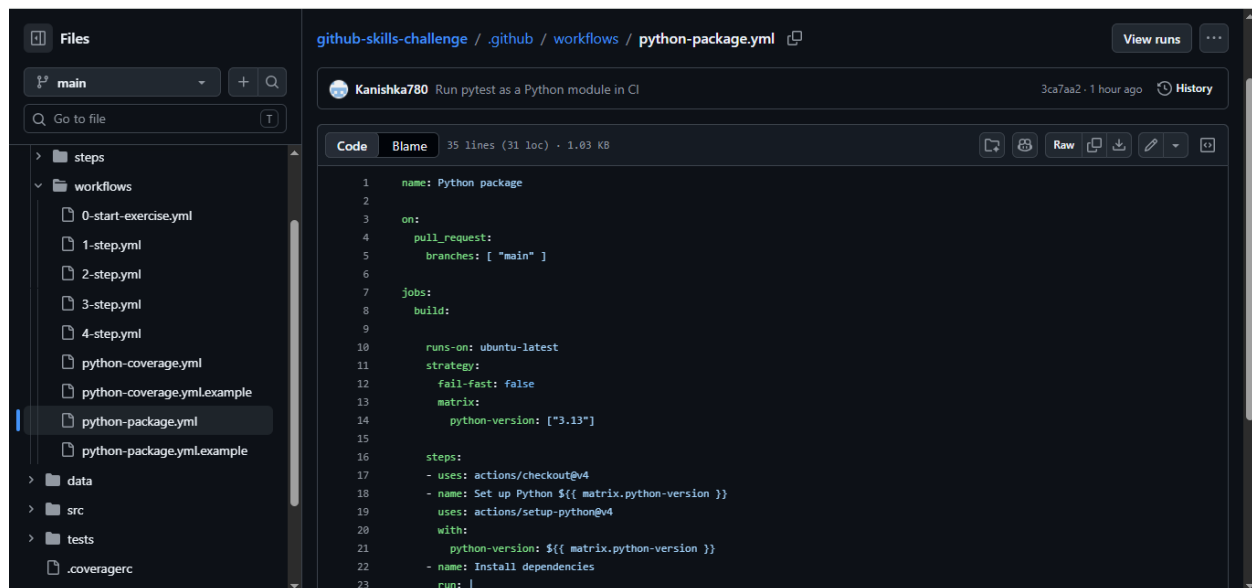
===== tests coverage =====
----- coverage: platform linux, python 3.13.15-final-0 -----

Name                               Stmts  Miss  Cover   Missing
-----
src/aiops_pipeline.py               37     29    22%    10-11, 15-38, 46-62
src/anomaly_detector.py             18      3    83%     21, 24, 28
src/calculations.py                  16      6    62%     11, 18, 24-27
src/event_consumer.py                 6      0   100%
src/event_producer.py                 9      1    89%     12
src/event_topic.py                  10      1    90%     15
-----
TOTAL                               96     40    58%
===== 8 passed in 0.12s =====

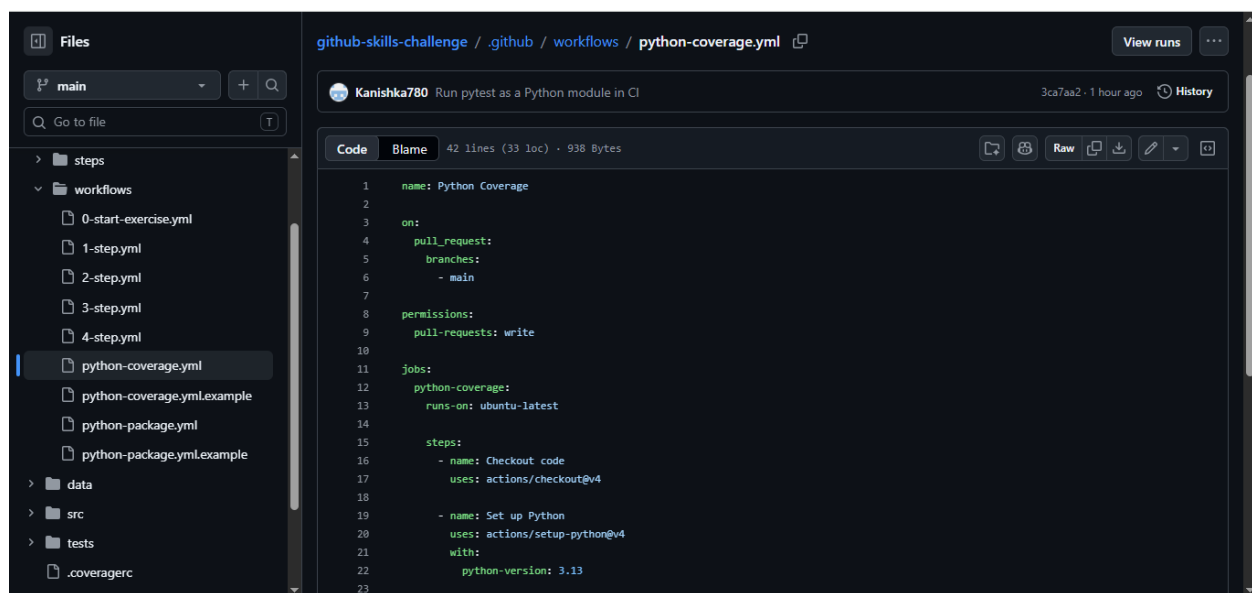
(.venv) @Kanishka780 →/workspaces/github-skills-challenge (main) $
```

5. Python Package Workflow

Heading: Python package GitHub Actions workflow



6. Coverage Workflow



Heading: Python coverage GitHub Actions workflow

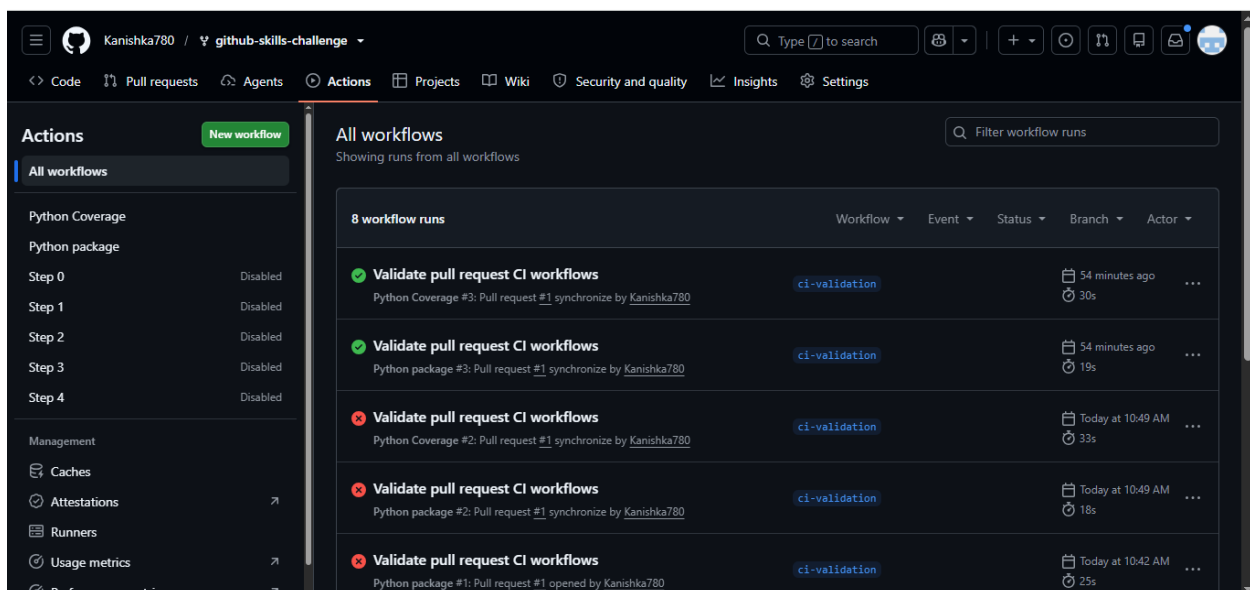
```
• (.venv) @Kanishka780 → /workspaces/github-skills-challenge (main) $ coverage report --fail-under=50
Name                               Stmts  Miss  Cover
-----
src/aiops_pipeline.py               37     29   22%
src/anomaly_detector.py             18      3   83%
src/calculations.py                 16      6   62%
src/event_consumer.py                6      0  100%
src/event_producer.py               9      1   89%
src/event_topic.py                  10      1   90%
-----
TOTAL                               96     40   58%
• (.venv) @Kanishka780 → /workspaces/github-skills-challenge (main) $
```

7. GitHub Actions Validation

Both workflows were successfully recognized and executed by GitHub Actions:

- Python package workflow: **Passed**
- Python coverage workflow: **Passed**

The workflows were triggered by pull request #1 targeting the `main` branch.



Heading: Successful GitHub Actions workflow runs

Use the screenshot showing green checks for:

- Python package
- Python Coverage

8. Pull Request

Summary

All jobs

python-coverage

Run details

Usage

Workflow file

Annotations

1 warning and 11 notices

python-coverage

succeeded 1 hour ago in 26s

Search logs

> Set up job1s

> Build py-cov-action/python-coverage-comment-action@v36s

> Checkout code0s

> Set up Python0s

> Install dependencies13s

> Run tests and generate coverage details1s

> Coverage comment3s

> Fail if below threshold0s

> Post Set up Python0s

> Post Checkout code0s

> Complete job0s

Python package

Validate pull request CI workflows #3

Re-run all jobs

...

Summary

All jobs

build (3.13)

Run details

Usage

Workflow file

Annotations

1 warning and 1 notice

build (3.13)

succeeded 1 hour ago in 16s

Search logs

> Set up job0s

> Run actions/checkout@v41s

> Set up Python 3.130s

> Install dependencies11s

> Lint with flake80s

> Test with pytest1s

> Post Set up Python 3.130s

> Post Run actions/checkout@v40s

> Complete job0s